



Optimization Homework 1

SF1811

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AI level 1: Used chatgpt to parse numerical results in to latex.

The intent of this homework assignment is to use the simplex method for solving a linear program in standard form,

$$\begin{aligned} \text{(LP)} \quad & \min_{x \in \mathbb{R}^n} c^T x \\ & \text{subject to } Ax = b, \\ & x \geq 0. \end{aligned}$$

There are Matlab functions `simplex.m` and `polyplot.m` available in Canvas to assist you.

Exercise 1

The simplex method is an iterative method. Its aim is to find the minimum of functions of the form $z(x) = C^T x$ with a constraint $Ax = b$. We'll illustrate it with the following pseudocode to keep it brief

Select the initial basic index tuple

We select linearly independent columns of A to form a new matrix A_β which we call the basic matrix. The basic index tuple $\beta = (\beta_1, \beta_2 \dots \beta_m)$ are the indices of the basic variables we have selected from x in to x_β corresponding to the column indices we have selected for the basic matrix. After this we define our non basic matrix and non basic variable A_ν and x_ν where the non basic index tuple $\nu = (\nu_1, \nu_2 \dots \nu_{n-m})$ are all the indices not included in β . Where we set $x_\nu = 0$.

loop

Step 1:

Solve $A_\beta \bar{b} = b$ for $\bar{b}$. This is solvable with multiplying with A_β^{-1} . This gives us the current basic variable solution $x_\beta = \bar{b}$.

Step 2:

Solve $A_\beta^T y = c_\beta$ for simplex multipliers y . (Similarly as before c_β are the corresponding coefficients to x_β)

Step 3:

Calculate $r_\nu^T = c_\nu^T - y^T A_\nu$ to find the reduced cost with respect to the non basic variables. Observe that this value is obtained from the objective value function $z = y^T b + (c_\nu^T - y^T A_\nu)x_\nu$. This measures the rate the objective value function improves if our choice of non-basic variables have been increased from zero.

Step 4: stopping criterion

if $r_\nu \geq 0$ **then**

then the basic feasible solution we found in this iteration is optimal and we can stop (any further increase to will increase the objective function).

return $x_\beta = \bar{b}$, $x_\nu = 0$

else

Step 5: Select Entering Variable

Choose index q such that $r_{\nu_q} < 0$. By dantzig's rule we choose q from the most negative component of r_ν .

Set $k = \nu_q$

▷ Variable x_k enters the basis

Step 6:

Solve $A_\beta \bar{a}_k = a_k$ for $\bar{a}_k$

Step 7: Ratio Test (Select Leaving Variable)

Since we want to ensure the feasibility condition $x_\beta \geq 0$ we want to know the maximum step size t_{max} . This tells us how much t can increase before a variable in x_β goes to zero. We want to remove the first variable that does this. Lets call the index of the leaving variable p . When calculating t_{max} we ensure $\bar{a}_{ik} > 0$ and identify index p :

$$t_{max} = \min_{i:\bar{a}_{ik}>0} \left(\frac{\bar{b}_i}{\bar{a}_{ik}} \right) = \frac{\bar{b}_p}{\bar{a}_{pk}}$$

Let ν_q and β_p change place

▷ Update Basis

end if

end loop

Exercise 2

Lucas's personal number was used to get the data from **hw1data**. Our matrices are given by

$$A = \begin{pmatrix} -0.9771 & 0.9951 & 1.0000 & 0 & 0 & 0 \\ -0.0085 & 0.9947 & 0 & 1.0000 & 0 & 0 \\ 1.9312 & 2.9730 & 0 & 0 & 1.0000 & 0 \\ 1.9425 & -1.0206 & 0 & 0 & 0 & 1.0000 \end{pmatrix}, b = \begin{pmatrix} 1.9526 \\ 2.9920 \\ 14.0105 \\ 6.0423 \end{pmatrix}, c = \begin{pmatrix} -1.0324 \\ -2.9633 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Where we have to solve the LP:

$$\begin{aligned}
& \min \quad z = -1.0324 x_1 - 2.9633 x_2 \\
& \text{subject to} \\
& \quad -0.9771 x_1 + 0.9951 x_2 + x_3 = 1.9526, \\
& \quad -0.0085 x_1 + 0.9947 x_2 + x_4 = 2.9920, \\
& \quad 1.9312 x_1 + 2.9730 x_2 + x_5 = 14.0105, \\
& \quad 1.9425 x_1 - 1.0206 x_2 + x_6 = 6.0423, \\
& \quad x_1, x_2, x_3, x_4, x_5, x_6 \geq 0.
\end{aligned}$$

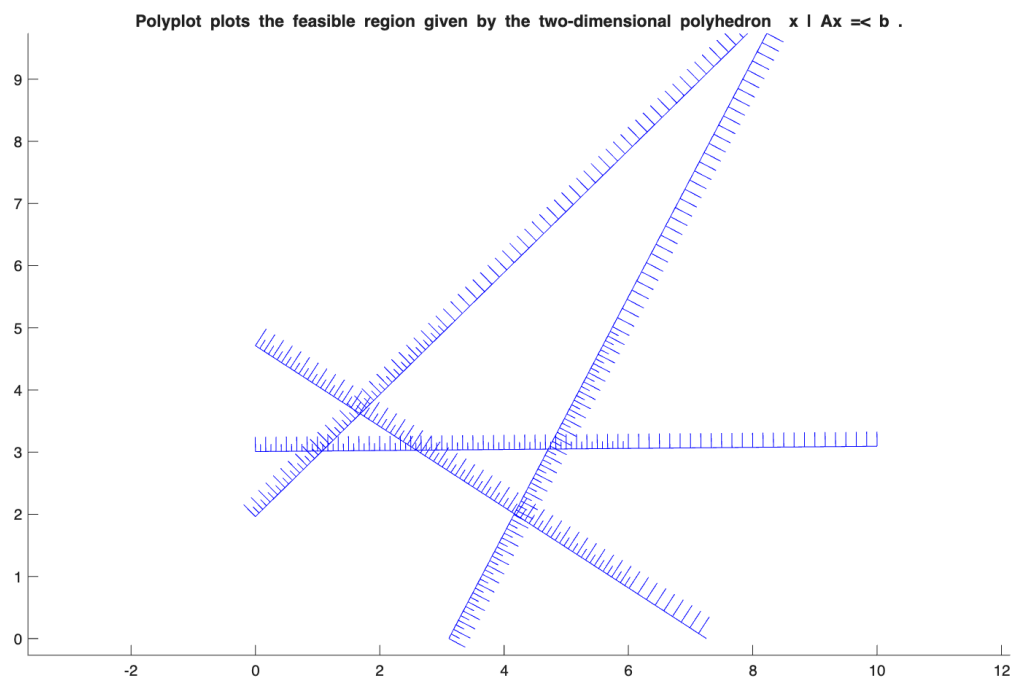


Figure 1: feasible region

Exercise 3

Iteration 1

Step 1: Basic Solution

We have that $A_\beta = I$ with linearly independent columns where the basic index tuple is $\beta = (3 \ 4 \ 5 \ 6)$

$$\text{We first solve } A_\beta \bar{b} = b \text{ for } \bar{b} \Leftrightarrow x_\beta = b = \begin{pmatrix} 1.9526 \\ 2.9920 \\ 14.0105 \\ 6.0423 \end{pmatrix} \Rightarrow x^{(1)} = \begin{pmatrix} 0 \\ 0 \\ 1.9526 \\ 2.9920 \\ 14.0105 \\ 6.0423 \end{pmatrix}$$

Step 2: Simplex Multipliers

$$\text{Solve } A_\beta^T y = c_\beta \text{ for } y \Leftrightarrow I^T y = y = c_\beta = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Step 3: Reduced Costs

$$\text{Calculate } r_\nu^T = c_\nu^T - y^T A_\nu = c_\nu^T - 0^T A_\nu = c_\nu = \begin{pmatrix} -1.0324 \\ -2.9633 \end{pmatrix}$$

Step 4: stopping criterion

We see that $r_\nu < 0$ so the BFS isn't optimal

Step 5: Select Entering Variable

Choose index $q = 2$ since $r_{\nu_2} < 0$ and $r_{\nu_2} = -2.96 = \min_{\nu_i \in \nu} r_{\nu_i}$ is the most negative value of r_ν .

So set $k = \nu_2$

▷ Variable x_2 will enter the basic variable and 2 will enter β

Step 6:

$$\text{Solve } A_\beta \bar{a}_k = a_k \text{ for } \bar{a}_k \Leftrightarrow I \bar{a}_k = \begin{pmatrix} 0.9951 \\ 0.9947 \\ 2.9730 \\ -1.0206 \end{pmatrix}$$

Step 7: Ratio Test (Select Leaving Variable)

Calculate t_{max} and identify index p :

$$t_{max} = \min_{i: \bar{a}_{ik} > 0} \left(\frac{\bar{b}_i}{\bar{a}_{ik}} \right) = \frac{\bar{b}_p}{\bar{a}_{pk}}$$

We can just do a linear search over every i to calculate this

$$t_{max} = \min_{a_{ik} > 0} \left\{ \frac{\bar{b}_1}{\bar{a}_{1k}}, \frac{\bar{b}_2}{\bar{a}_{2k}}, \frac{\bar{b}_3}{\bar{a}_{3k}}, \frac{\bar{b}_4}{\bar{a}_{4k}} \right\} = \min_{a_{ik} > 0} \{1.9621, 3.0080, 4.7126, -5.9205\} = 1.9621$$

. So the index leaving is $\beta_p = \beta_1$.

Let $\nu_2 = 2$ and $\beta_1 = 3$ change place in **iteration 2**

▷ Update Basis

Iteration 2

$$\begin{aligned}x^{(2)} &= \begin{pmatrix} 0 \\ 1.9621 \\ 0 \\ 1.0403 \\ 8.1771 \\ 8.0447 \end{pmatrix} \\r_{\nu}^{(2)} &= \begin{pmatrix} -3.9421 \\ 2.9778 \end{pmatrix} \\y^{(2)} &= \begin{pmatrix} -2.9778 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\\beta &= (2 \quad 4 \quad 5 \quad 6)\end{aligned}$$

Iteration 3

$$\begin{aligned}x^{(3)} &= \begin{pmatrix} 1.0745 \\ 3.0172 \\ 0 \\ 0 \\ 2.9652 \\ 7.0343 \end{pmatrix} \\r_{\nu}^{(3)} &= \begin{pmatrix} 4.0716 \\ -1.0919 \end{pmatrix} \\y^{(3)} &= \begin{pmatrix} 1.0919 \\ -4.0716 \\ 0 \\ 0 \end{pmatrix} \\\beta &= (2 \quad 1 \quad 5 \quad 6)\end{aligned}$$

Iteration 4

$$x^* = \begin{pmatrix} 2.5900 \\ 3.0302 \\ 1.4680 \\ 0 \\ 0 \\ 4.1036 \end{pmatrix}$$

$$r_\nu^* = \begin{pmatrix} 1.3634 \\ 0.5406 \end{pmatrix}$$

$$y^* = \begin{pmatrix} 0 \\ -1.3634 \\ -0.5406 \\ 0 \end{pmatrix}$$

$$\beta = (2 \quad 1 \quad 3 \quad 6)$$

All reduced costs are $r_\nu \geq 0$ so we found the optimal solution.

Optimal BFS

$$x^* = \begin{pmatrix} 2.5900 \\ 3.0302 \\ 1.4680 \\ 0 \\ 0 \\ 4.1036 \end{pmatrix}$$

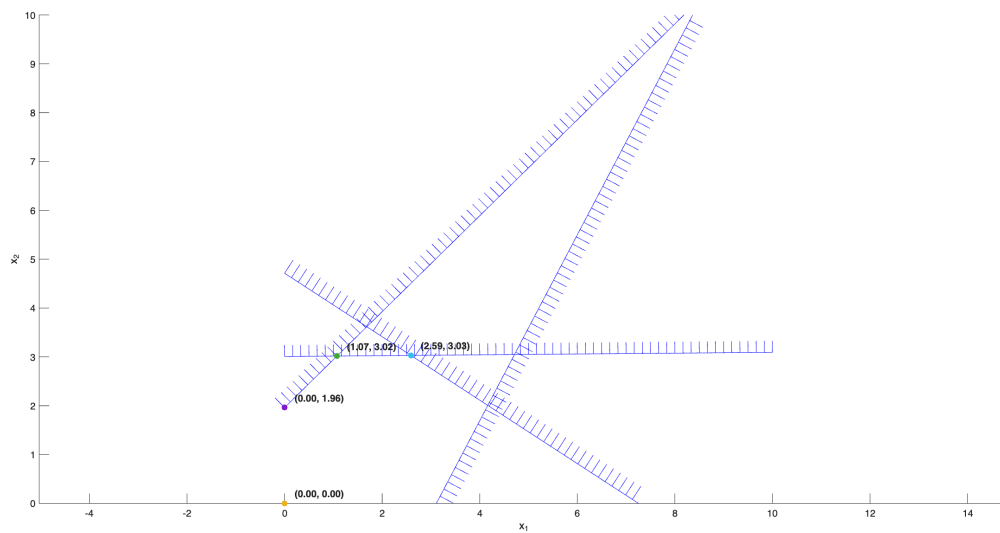


Figure 2: feasible region with iteration points included

Exercise 4

We'll repeat exercise 3 but with the choice of $q = 1$ in **iteration 1**. This is still valid since $r_{\nu_1} = -1.0324$ is non positive. Recall that $z = \bar{z} + r_{\nu}x_{\nu}$. If x_1 becomes a non basic variable it still ensures the objective value function $z = c^T x$ decreases (just less than if we chose x_2 first).

Iteration 1

$$\begin{aligned}x^{(1)} &= \begin{pmatrix} 0 \\ 0 \\ 1.9526 \\ 2.9920 \\ 14.0105 \\ 6.0423 \end{pmatrix} \\r_{\nu}^{(1)} &= \begin{pmatrix} -1.0324 \\ -2.9633 \end{pmatrix} \\y^{(1)} &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\\beta &= (3 \quad 4 \quad 5 \quad 6)\end{aligned}$$

Iteration 2

$$\begin{aligned}x^{(2)} &= \begin{pmatrix} 3.1105 \\ 0 \\ 4.9920 \\ 3.0184 \\ 8.0035 \\ 0 \end{pmatrix} \\r_{\nu}^{(2)} &= \begin{pmatrix} 0.5315 \\ -3.5057 \end{pmatrix} \\y^{(2)} &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ -0.5315 \end{pmatrix} \\\beta &= (3 \quad 4 \quad 5 \quad 1)\end{aligned}$$

Iteration 3

$$\begin{aligned}x^{(3)} &= \begin{pmatrix} 4.1650 \\ 2.0071 \\ 4.0251 \\ 1.0310 \\ 0 \\ 0 \end{pmatrix} \\r_{\nu}^{(3)} &= \begin{pmatrix} -0.3425 \\ 0.8791 \end{pmatrix} \\y^{(3)} &= \begin{pmatrix} 0 \\ 0 \\ -0.8791 \\ 0.3425 \end{pmatrix} \\\beta &= (3 \quad 4 \quad 2 \quad 1)\end{aligned}$$

Iteration 4

$$\begin{aligned}x^* &= \begin{pmatrix} 2.5900 \\ 3.0302 \\ 1.4680 \\ 0 \\ 0 \\ 4.1036 \end{pmatrix} \\r_{\nu}^* &= \begin{pmatrix} 1.3634 \\ 0.5406 \end{pmatrix} \\y^* &= \begin{pmatrix} 0 \\ -1.3634 \\ -0.5406 \\ 0 \end{pmatrix} \\\beta &= (3 \quad 6 \quad 2 \quad 1)\end{aligned}$$

All reduced costs are $r_{\nu} \geq 0$, optimal solution found.

Optimal BFS

$$x^* = \begin{pmatrix} 2.5900 \\ 3.0302 \\ 1.4680 \\ 0 \\ 0 \\ 4.1036 \end{pmatrix}$$

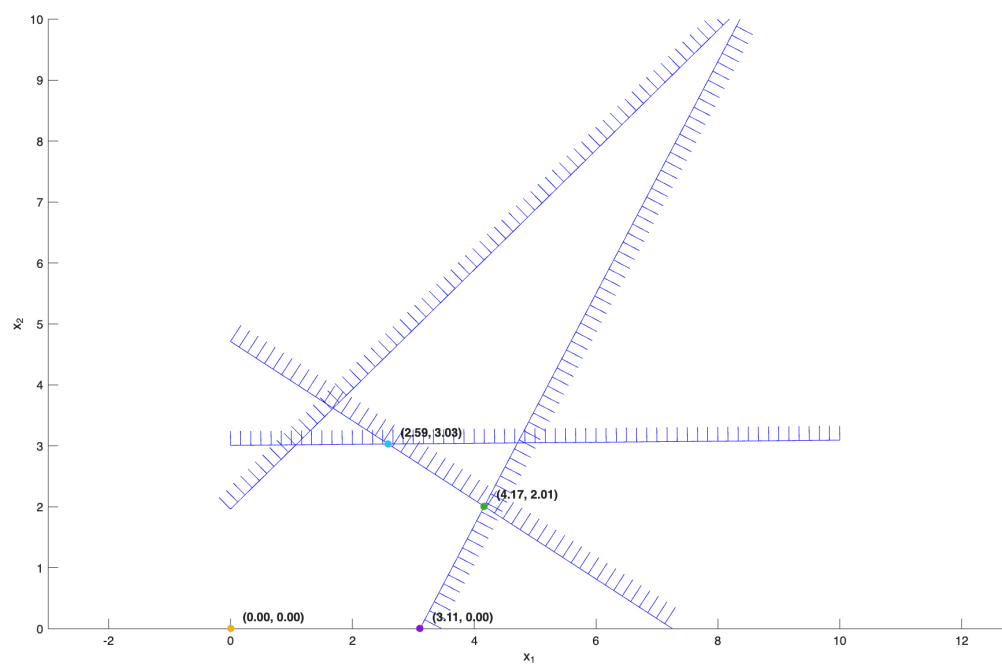


Figure 3: feasible region with iteration points included, but with choice of $q = 1$ in `it1`